

Jeremy Fontenot's On-Premises Home Lab Documentation

Professional APA 7 Technical Portfolio Paper

Prepared by Jeremy Fontenot

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Executive Abstract

This professional paper documents Jeremy Fontenot's personal on-premises home lab as a technical portfolio artifact for recruiters, hiring managers, systems administrators, infrastructure engineers, and technical interviewers. It summarizes architecture, implementation, administration, validation, troubleshooting, automation, backup, recovery, evidence governance, and known limitations using current repository-backed evidence.

The lab evidence supports a career direction from experienced Service Desk and IT Support into systems administration and infrastructure operations. It does not represent employer production administration, client infrastructure ownership, enterprise-scale operations, or guaranteed availability.

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Purpose and Scope

The purpose of this document is to present a public-safe, evidence-based account of a personal home lab used for systems administration practice. The scope includes architecture, implementation, validation, documentation, and operational boundaries.

Professional and Technical Context

Jeremy is positioned as an experienced Service Desk and IT Support professional progressing toward systems administration and infrastructure operations. The home lab demonstrates technical initiative and disciplined documentation without converting personal lab activity into employment claims.

Evidence and Validation Methodology

Evidence quality is established through commands, authenticated inventories, structured output, manifests, hashes, configuration state, testing results, source records, and traceable provenance. Dates are retained as metadata, not as proof by themselves.

Evidence Classification Model

Claims are classified as Planned, Configured, Implemented, Tested, Validated, Historical, Limitation, or Inconclusive. Public copy does not upgrade configured or tested behavior into validated behavior unless direct evidence supports that classification.

Physical Infrastructure

The current supported physical host is a Dell PowerEdge R710 used as a personal nonproduction lab host.

Virtualization Platform

Current source records support Proxmox VE 9.2.3 with VMs 100, 200, 300, and 400 running. VM 100 is pfSense, VM 200 is DC01, VM 300 is WS01, and VM 400 is Linux01.

Storage Architecture

Current records support backup storage named backup-hdd at /mnt/pve/backup-hdd. Historical references to other backup storage labels are not treated as current unless future evidence supports them.

Network Architecture

The lab uses pfSense, internal lab networks, VLAN 30 for Linux01, and an OpenVPN management path documented in the operations records. Linux01 current supported placement is 10.10.30.20/24.

pfSense Routing and Segmentation

pfSense is represented as VM 100. Public claims distinguish current configuration records from behavior that was not independently tested.

VLAN Architecture

Linux01 is currently treated as VLAN 30 at 10.10.30.20/24. Older Linux01 10.10.20.x records are historical setup or pre-reconciliation evidence.

VPN and Management Path

Route and TCP evidence support a management path for selected services. Timeout records remain timeout observations and are not converted into outage claims.

Windows Server and Active Directory

DC01 is documented as a Windows Server 2022 domain controller for ad.jeremyfontenot.online.

Retained evidence supports AD DS, DNS, DHCP, Group Policy, and FSMO role review where cited.

DNS

DNS evidence includes AD-integrated DNS records and forward/reverse lookup validation in retained public screenshots and text artifacts.

DHCP

DHCP evidence includes scope, options, statistics, and redacted lease review records. Sensitive client identifiers remain redacted.

Group Policy

Group Policy evidence supports GPO inventory and domain or organizational-unit link review.

File Services and Permissions

File-service evidence is treated as tested only where specific permission validation records exist. No broader enterprise file-service assurance is claimed.

WS01 Workstation Architecture and Validation

WS01 is documented as a Windows domain workstation. Current records include authenticated inventory evidence, while guest-agent limitations are kept visible where applicable.

Linux01 Architecture and Network Placement

Linux01 is documented as Ubuntu Server 26.04 LTS on VLAN 30 at 10.10.30.20/24.

Linux Active Directory Integration

Linux01 evidence supports domain integration through realm membership, identity resolution, and related validation artifacts.

SSSD and Kerberos

The Linux01 SSSD remediation record documents a responder activation conflict, configuration correction, service restart, failed-unit cleanup, identity lookup, and a new AD-authenticated SSH session.

SSH, UFW, sudo, and Linux Security Controls

Linux01 records support SSH, UFW, delegated sudo, and related Linux security-control claims where linked.

Centralized Logging

Centralized logging is described within the boundaries of available pfSense and operations documentation. Full security monitoring maturity is not claimed.

QEMU Guest Agent and Maintenance State

Linux01 records include QEMU guest-agent response and maintenance-state evidence. WS01 guest-agent limitations remain visible.

Backup Architecture

Current inventory supports an enabled scheduled backup job named backup-critical-lab-vms for VMs 100, 200, 300, and 400 using snapshot mode and zstd compression.

Recovery Validation

Retained documentation records isolated startup restore checks for temporary WS01 and Linux01 restore VMs. The current records do not establish recurring disaster-recovery assurance, RTO, or RPO.

Storage Monitoring

Storage monitoring claims are limited to available inventory and utilization records.

Read-Only Remote Validation

Read-only validation includes repository checks, Proxmox inventory, route evidence, DNS checks, and bounded TCP reachability. It does not prove full service health for every endpoint.

Repository Validation

The operations repository records validation scripts, inventories, evidence summaries, and source-of-truth reconciliation. The website repository publishes sanitized reviewer-safe summaries.

Evidence Integrity

Public artifacts are mapped through manifests, SHA-256 hashes, source paths, and provenance records.

Screenshot Evidence Review

Screenshots are used only where they support adjacent claims, with full-size originals linked from previews and captions scoped to visible evidence.

Unresolved Investigations and Known Limitations

Known limitations include personal lab scope, no production administration claim, no enterprise scale claim, no guaranteed uptime, no full security assurance, and no recurring restore assurance.

Table 1

Evidence Classification Summary

Classification	Use	Boundary
Validated	Direct evidence confirms the stated behavior or state.	Used only where source artifacts support the claim.
Tested	A relevant test produced a recorded result.	Does not imply complete service assurance.
Configured	Configuration is present.	Behavior may require separate testing.
Limitation	Known boundary, gap, or unsupported conclusion.	Kept visible for reviewers.

Figure 1

Current supported home-lab architecture summary. Source note: derived from current operations repository source-of-truth and reconciliation records.

References

Fontenot, J. (2026). Home lab source-of-truth repository records. fontenotjeremy71-hub/jeremy-homelab-ops, commit 3d42a4808a6a0607d78f4f4e7fd1e71783729a6b.

Fontenot, J. (2026). Public evidence library and portfolio repository.
fontenotjeremy71-hub/jeremyfontenot, commit
a95259d90717695acebf8840d740fde178adda8a.

American Psychological Association. (2020). Publication manual of the American Psychological Association (7th ed.).

Appendix A: Source Artifacts

home-lab-source-of-truth.md

home-lab-source-reconciliation.md

home-lab-inventory-validation.md

linux01-sssd-remediation.md

backup-and-restore-validation.md

Appendix B: Claim Boundaries

The document intentionally avoids claims of production administration, client infrastructure ownership, enterprise scale, guaranteed uptime, unsupported business impact, full security assurance, or recurring restore assurance.